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MICRO PROJECT SYNOPSIS

ON

Library Management System

Submitted for the Subject

DATABASE MANAGEMENT SYSTEMS (DBMS)

Submitted By

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Formatting Specifications(Not to be printed, only for reference)

Specification	Requirement
Paper Size	A4
Font Family	Times New Roman
Body Text	12 pt, Regular
Main Headings	14 pt, Bold
Subheadings	12 pt, Bold
Project Title	16 pt, Bold, Center aligned
Cover Page Main Heading	18 pt, Bold, Center aligned
Line Spacing	1.5
Paragraph Alignment	Justified
Margins	Top: 0.75", Bottom: 0.75", Left: 1", Right: 1"
Page Number	Bottom-center
Heading Numbering	1, 2, 3…; subheadings 8.1, 8.2…
Tables	Times New Roman, 11–12 pt; captions above tables
Figures/Diagrams	Centered; caption below figure
Printing	A4, single-sided print, unless instructed otherwise

Important: Students should edit only the placeholders and project-specific content. Maintain the same formatting throughout the report.

Table of Content

1. Project Title

Library Management System

2. Introduction

A Library Management System is a web-based application that helps manage the daily activities of a library. It stores information about books, authors, categories, and members in a database. It also helps keep track of issued books, returned books, overdue books, and fines.

The main purpose of this project is to make library management easier, faster, and more organized. Using HTML, CSS, JavaScript, Python Flask, and PostgreSQL, the system reduces manual work and makes it easier to access and update library records.

3. Problem Statement

Managing a library involves maintaining records of books, authors, categories, members, and book borrowing activities. When these records are handled manually, it can become difficult to find books, check their availability, update member details, and keep track of issued and returned books. Manual record-keeping may also lead to duplicate entries, missing information, and errors.

Another challenge is tracking book issue dates, due dates, overdue books, and fines. Without a proper system, librarians may find it difficult to identify overdue books and maintain accurate records of library transactions. Managing relationships between books, authors, members, and loan records can also become complicated when the information is not properly organized.

To solve these problems, a Library Management System is proposed using HTML, CSS, JavaScript, Python Flask, and PostgreSQL. The system will provide a centralized database to store, retrieve, and update library information efficiently. It will also use important DBMS

concepts such as primary keys, foreign keys, joins, nested subqueries, aggregate functions, triggers, views, indexes, and transactions to maintain data accuracy and simplify database operations. The main aim is to reduce manual work, minimize errors, and make library management easier, faster, and more organized.

4. Objectives

- To maintain book, author, category, and member records.
- To manage book issuing and returning.
- To track overdue books and calculate fines.
- To reduce manual work and avoid duplicate records.
- To apply DBMS concepts such as joins, subqueries, functions, triggers, views, indexes, and transactions.
- To provide a simple and user-friendly interface for library management.

5. Scope of the Project

- Book management: Add, view, update, and delete book records.
- Member management: Store and manage library member details.
- Issue and return: Record book lending and return activities.
- Overdue tracking: Identify overdue books and display applicable fines.
- Database operations: Perform searching, filtering, sorting, and report generation.
- Data integrity: Use primary keys, foreign keys, and constraints to maintain correct records.
- Database analysis: Generate useful information using SQL queries and aggregate functions.

6. Proposed System

The proposed system is a web-based application with a frontend developed using HTML, CSS, and JavaScript, a backend developed using Python Flask, and PostgreSQL as the database.

Users can manage books and members, issue and return books, and check overdue records through the application. The backend processes user requests and communicates with the PostgreSQL database to store and retrieve information.

The system also uses advanced SQL concepts, including joins, subqueries, aggregate functions, PL/pgSQL functions, triggers, views, indexes, and transactions. These features help maintain data accuracy and make library operations more efficient.

7. Database Design

7.1 Entities and Attributes

1. Authors

- author_id
- first_name
- last_name
- email

2. Categories

- category_id
- category_name
- description

3. Books

- book_id

- isbn
- title
- category_id
- publisher
- publication_year
- total_copies
- available_copies

4. Book_Authors

- book_id
- author_id

5. Members

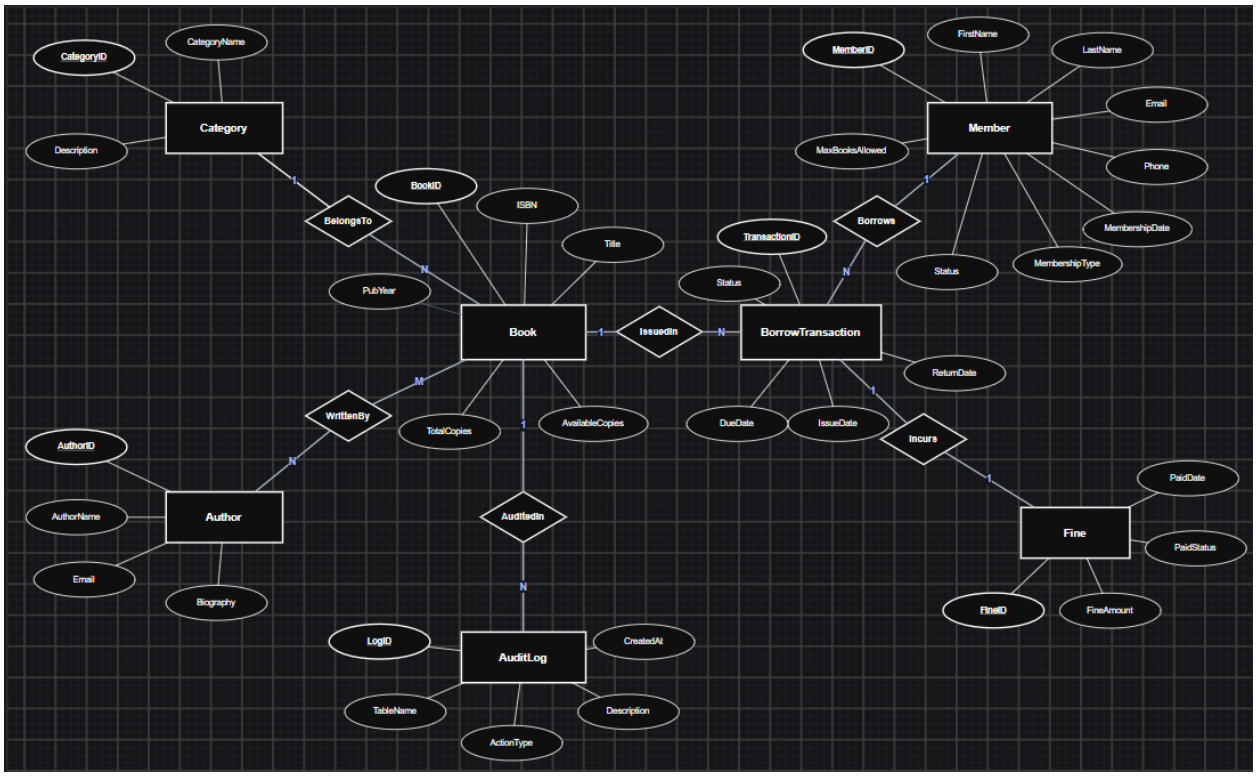
- member_id
- member_code
- first_name
- last_name
- email
- phone
- address
- membership_type
- status
- joined_date

6. Loans

- loan_id
- book_id
- member_id

- issue_date
- due_date
- return_date
- fine_amount
- status

7.2 ER/EER Diagram



7.3 Relational Schema

- Authors: Authors(author_id PK, first_name, last_name, email)
- Categories: Categories(category_id PK, category_name, description)
- Books: Books(book_id PK, isbn, title, category_id FK, publisher, publication_year, total_copies, available_copies)
- Book_Authors: Book_Authors(book_id PK, FK, author_id PK, FK)

- Members: Members(member_id PK, member_code, first_name, last_name, email, phone, address, membership_type, status, joined_date)
- Loans: Loans(loan_id PK, book_id FK, member_id FK, issue_date, due_date, return_date, fine_amount, status)

7.4 Primary Keys and Foreign Keys

Primary Keys (PK)

- Authors: author_id
- Categories: category_id
- Books: book_id
- Book_Authors: Composite key (book_id, author_id)
- Members: member_id
- Loans: loan_id

8. SQL Implementation

- DDL (Data Definition Language): CREATE TABLE and ALTER TABLE to create and modify database tables.
- DML (Data Manipulation Language): INSERT, UPDATE, and DELETE to add, modify, and remove records.
- Basic SQL queries: SELECT, WHERE, ORDER BY, and DISTINCT to retrieve and filter information.
- Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and CHECK to maintain data accuracy.
- Joins: INNER JOIN and LEFT JOIN to retrieve related information from multiple tables.
- Nested subqueries: To find records based on the results of another query.
- Aggregate functions: COUNT(), SUM(), AVG(), MIN(), and MAX() to summarize library data.
- GROUP BY and HAVING: To group records and filter grouped results.

- Functions and triggers: PL/pgSQL functions and triggers to automate tasks, such as updating available book copies during issue and return operations.
- Views and indexes: To simplify frequently used queries and improve data retrieval.
- Transactions: BEGIN, COMMIT, and ROLLBACK to ensure related database operations are completed safely.

9. Sample SQL Queries

```
CREATE TABLE books (
    book_id SERIAL PRIMARY KEY,
    isbn VARCHAR(20) NOT NULL UNIQUE,
    title VARCHAR(255) NOT NULL,
    category_id INT REFERENCES categories(category_id) ON DELETE SET NULL ON UPDATE CASCADE,
    publication_year INT CHECK (publication_year >= 1000 AND publication_year <= 2100),
    total_copies INT NOT NULL DEFAULT 1 CHECK (total_copies >= 0),
    available_copies INT NOT NULL DEFAULT 1 CHECK (available_copies >= 0 AND available_copies <= total_copies),
    price NUMERIC(8, 2) DEFAULT 0.00 CHECK (price >= 0),
    rack_location VARCHAR(50),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
SELECT * FROM books;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	book_id [PK] integer	isbn character varying (20)	title character varying (255)	category_id integer	publication_year integer	total_copies integer	available_copies integer	pric num
1	4	978-0262033848	Introduction to Algorithms (CLRS 3rd Ed)	1	2009	10	8	
2	5	978-0451524935	1984	3	1949	12	10	
3	6	978-0553293357	Foundation	4	1951	7	4	
4	7	978-0345539434	Cosmos	6	1980	6	5	
5	8	978-0385490818	The Handmaid's Tale	3	1985	8	6	
6	9	978-0201558029	Concrete Mathematics: A Foundation for Computer Scie...	5	1994	4	3	
7	10	978-0131103627	The C Programming Language (2nd Ed)	1	1988	5	4	
8	1	978-0201896831	The Art of Computer Programming, Vol 1	1	1997	5	3	
9	2	978-0132350884	Clean Code: A Handbook of Agile Software Craftsmanship	1	2008	8	5	
10	3	978-0136042594	Artificial Intelligence: A Modern Approach (4th Ed)	2	2020	6	4	

```
INSERT INTO members (member_code, first_name, last_name, email, phone, address, membership_type, status, joined_date) VALUES
('MEM-1001', 'Alice', 'Smith', 'alice.smith@university.edu', '+1-555-0101', '124 Science Hall, Cambridge, MA', 'Faculty', 'Active',
('MEM-1002', 'Bob', 'Johnson', 'bob.j@college.edu', '+1-555-0102', '55 Dormitory Row, Boston, MA', 'Student', 'Active', '2025-02-01'
('MEM-1003', 'Charlie', 'Brown', 'charlie.brown@peanuts.net', '+1-555-0103', '742 Evergreen Terrace, Springfield', 'Standard', 'Acti
('MEM-1004', 'Diana', 'Prince', 'diana.prince@themyscira.gov', '+1-555-0104', 'Gate 4, Embassy Row, Washington, DC', 'Premium', 'Act
('MEM-1005', 'Evan', 'Wright', 'evan.wright@media.org', '+1-555-0105', '88 Sunset Blvd, Los Angeles, CA', 'Standard', 'Suspended', '
('MEM-1006', 'Fiona', 'Gallagher', 'fiona.g@southside.org', '+1-555-0106', '2119 North Wallace St, Chicago, IL', 'Student', 'Active'
('MEM-1007', 'George', 'Clark', 'george.clark@techcorp.io', NULL, '10 Tech Park Way, Austin, TX', 'Standard', 'Expired', '2024-01-10
SELECT * FROM members;
```

```sql
SELECT
    b.book_id,
    b.title,
    b.isbn,
    c.name AS category_name,
    b.price
FROM books b
INNER JOIN categories c ON b.category_id = c.category_id
ORDER BY b.title ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	book_id integer	title character varying (255)	isbn character varying (20)	category_name character varying (100)	price numeric (8,2)
1	5	1984	978-0451524935	Literature & Fiction	15.99
2	3	Artificial Intelligence: A Modern Approach (4th Ed)	978-0136042594	Artificial Intelligence	89.99
3	2	Clean Code: A Handbook of Agile Software Craftsmanship	978-0132350884	Computer Science	44.95
4	9	Concrete Mathematics: A Foundation for Computer Scie…	978-0201558029	Mathematics	59.99
5	7	Cosmos	978-0345539434	Physics & Astronomy	22.00
6	6	Foundation	978-0553293357	Science Fiction	17.50
7	4	Introduction to Algorithms (CLRS 3rd Ed)	978-0262033848	Computer Science	79.50
8	1	The Art of Computer Programming, Vol 1	978-0201896831	Computer Science	69.99
9	10	The C Programming Language (2nd Ed)	978-0131103627	Computer Science	49.00
10	8	The Handmaid's Tale	978-0385490818	Literature & Fiction	16.95

Query | Query History

```sql
SELECT
    title,
    price,
    ROUND((SELECT AVG(price) FROM books), 2) AS library_avg_price,
    ROUND(price - (SELECT AVG(price) FROM books), 2) AS diff_from_average
FROM books
WHERE price > (SELECT AVG(price) FROM books)
ORDER BY price DESC;
```

Data Output | Messages | Notifications

Showing rows:

	title character varying (255)	price numeric (8,2)	library_avg_price numeric	diff_from_average numeric
1	Artificial Intelligence: A Modern Approach (4th Ed)	89.99	46.59	43.40
2	Introduction to Algorithms (CLRS 3rd Ed)	79.50	46.59	32.91
3	The Art of Computer Programming, Vol 1	69.99	46.59	23.40
4	Concrete Mathematics: A Foundation for Computer Scie...	59.99	46.59	13.40
5	The C Programming Language (2nd Ed)	49.00	46.59	2.41

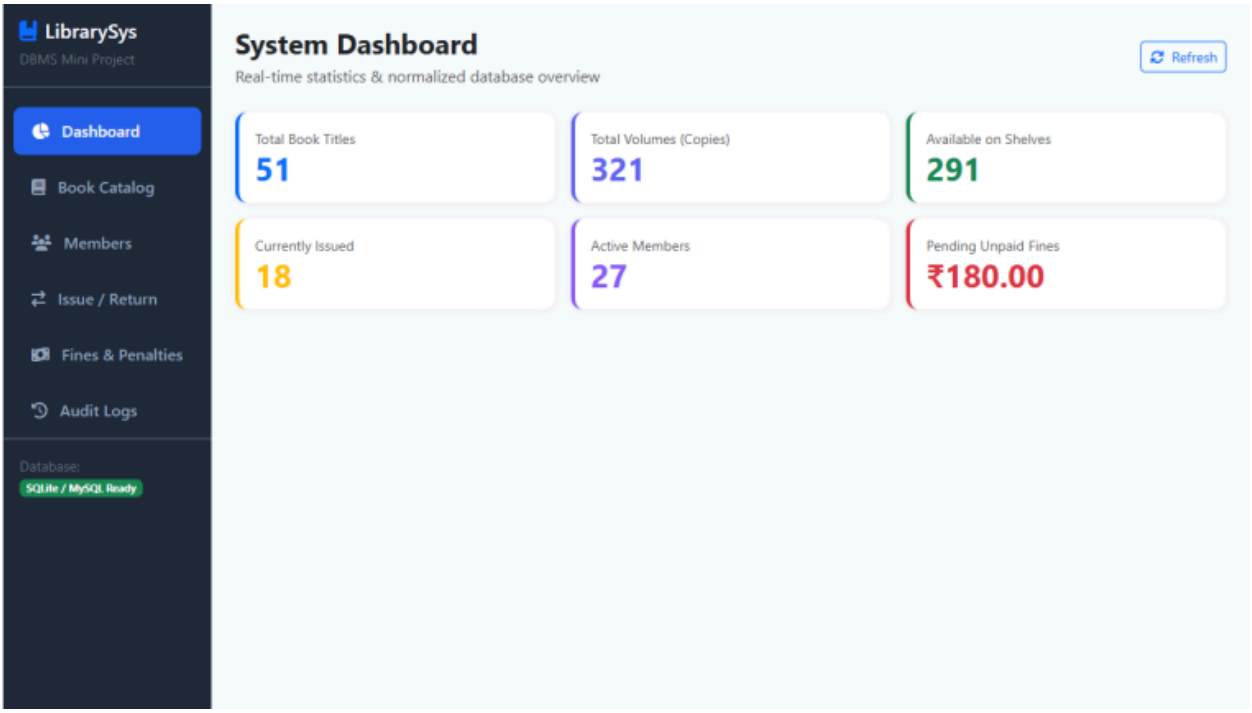
```sql
SELECT loan_id, book_title, borrower_name, issue_date, due_date, status, estimated_fine
FROM vw_active_loans;
```

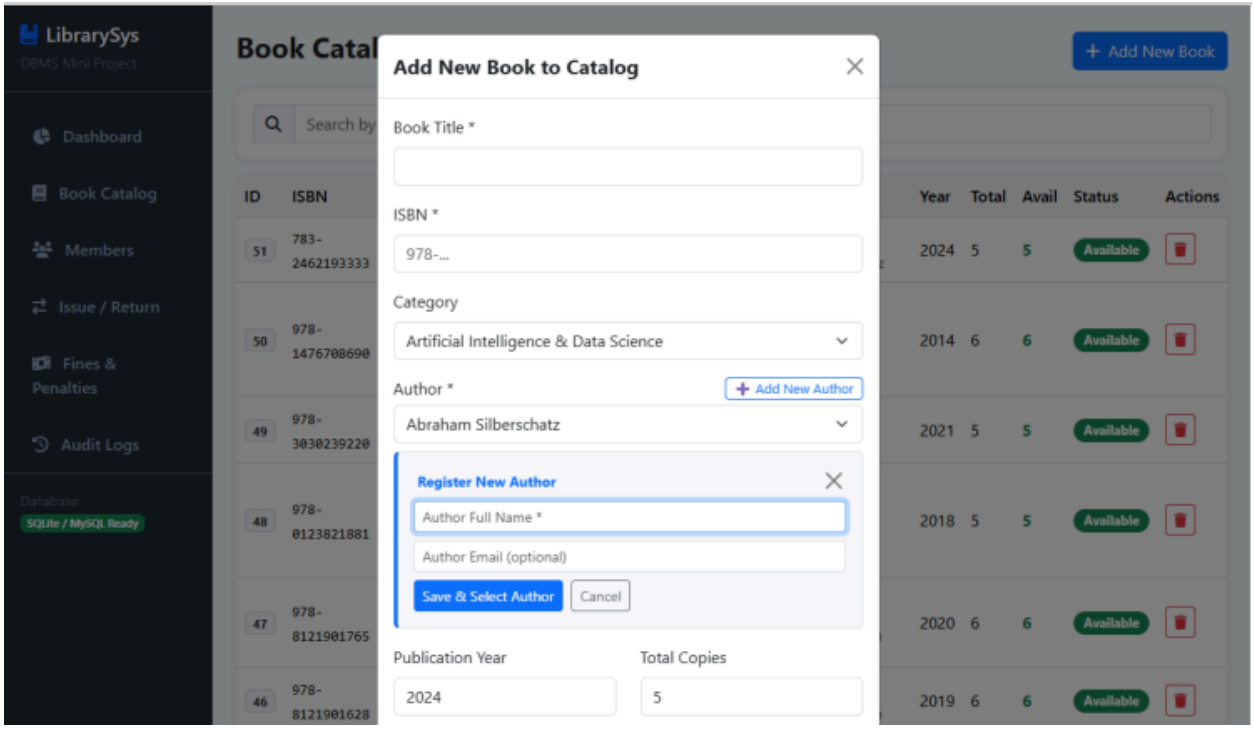
Data Output | Messages | Notifications

Showing rows: 1 to 11 Page No: 1

	loan_id integer	book_title character varying (255)	borrower_name text	issue_date date	due_date date	status character varying (20)	estimated_fine numeric
1	3	The Art of Computer Programming, Vol 1	Diana Prince	2026-10-21	2026-11-04	Issued	0.00
2	4	Clean Code: A Handbook of Agile Software Craftsmans...	Fiona Gallagher	2026-10-19	2026-11-02	Issued	0.00
3	5	Clean Code: A Handbook of Agile Software Craftsmans...	Charlie Brown	2026-10-23	2026-11-06	Issued	0.00
4	6	Artificial Intelligence: A Modern Approach (4th Ed)	Alice Smith	2026-10-22	2026-11-05	Issued	0.00
5	7	Introduction to Algorithms (CLRS 3rd Ed)	Bob Johnson	2026-10-01	2026-10-15	Overdue	12.00
6	8	Introduction to Algorithms (CLRS 3rd Ed)	Charlie Brown	2026-10-24	2026-11-07	Issued	0.00
7	9	1984	Evan Wright	2026-10-06	2026-10-20	Overdue	7.00
8	10	1984	Fiona Gallagher	2026-10-18	2026-11-01	Issued	0.00
9	11	Foundation	Diana Prince	2026-10-20	2026-11-03	Issued	0.00
10	12	Foundation	Bob Johnson	2026-10-08	2026-10-22	Overdue	5.00
11	13	The Handmaid's Tale	Alice Smith	2026-10-24	2026-11-07	Issued	0.00

10. Expected Outcome





11. Conclusion

The Library Management System helps make library work easier and more organized by managing book details, member information, and book issue and return records in one place. It reduces the need for manual record-keeping and helps avoid common mistakes such as duplicate entries and incorrect information. By using HTML, CSS, JavaScript, Python Flask, and PostgreSQL, we have developed a system that makes storing, updating, and finding information more convenient.

Through this project, we also gained practical knowledge of important DBMS concepts such as keys, constraints, joins, subqueries, aggregate functions, views, indexes, transactions, functions, and triggers. It helped us understand how a database works in a real-world application and how different technologies can be connected to build a useful system. In the future, more features like barcode scanning, due-date reminders, and online book reservations can be added to make the system even more useful.

12. References

1. PostgreSQL Global Development Group, *PostgreSQL Documentation*. Available at: https://www.postgresql.org/docs/
2. Python Software Foundation, Python Documentation. Available at: https://docs.python.org/3/
3. Pallets Projects, Flask Documentation. Available at: https://flask.palletsprojects.com/
4. MDN Web Docs, HTML, CSS, and JavaScript Guides. Available at: https://developer.mozilla.org/
5. W3Schools, SQL Tutorial. Available at: https://www.w3schools.com/sql/
6. Elmasri, R. and Navathe, S. B., Fundamentals of Database Systems. Pearson Education.